

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Supplementary Summer Examination – 2025****Course: B Tech.****Branch: Civil****Semester: VI****Subject Code & Name: BTCVC606_Y18 (Building Planning & Design)****Max Marks: 60****Date:12/07/25****Duration: 3 Hr.****Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Art of adding aesthetics to spaces is called ____ a.design b. planning c. interior d. acoustics	CO 1	1
2	Position of door & window in internal spaces for better light & ventilation is called ____ a.ventilation b. aspect c. prospect d. interior	CO 1	1
3	Horizontal angle of sun along the horizon is called ____? a. Azimuth b. Altitude c. acute angle d. sun angle	CO 1	1
4	Which rating system is used for green building in India? a. BREEAM b. LEED® c. GRIHA d. CASBEE	CO 2	1
5	Arrangement of functional spaces as per priority and interdependency is called ____? a. Aspect b. Prospect c. Grouping d. Privacy	CO 2	1
6	Which trap is provided at the junction of house drain and street sewer? a. Nahni trap b. Gully trap c. Intercepting d. Q trap	CO 3	1
7	What is the diameter of soil water drain pipe? a.100 mm b. 75 mm c. 50mm d. 40mm	CO 3	1
8	Which method is used for lighting design in building?	CO 4	1

	a. Lighting method	b. Lumen method	c. Lux method	d. wiring method		
9	Which part is used in air conditioning for dust control?				CO 4	1
	a. exhaust fan	b. compressor	c. damper	d. condenser		
10	Travel distance for staircase exit in residential building shall be ____				CO 5	1
	a. 22.5 m	b. 30 m	c. 35 m	d. 40 m		
11	Function of artificial ventilation is to provide?				CO 5	1
	a. aesthetics	b. air circulation	c. cool temperature	d. light		
12	Science of managing sound inside any enclosed space is called ____?				CO 5	1
	a. sound engineering	b. noise control	c. acoustics	d. civil engineering		
Q. 2	Solve the following.				CO 1	12
A)	Describe any three principals of Planning?					6
B)	What are the bye laws for size, area & height of WC & bath?					6
Q.3	Solve the following.				CO 2	12
A)	What are the requirements of municipal drawing?					6
B)	Which bye laws are used for residential building?					6
Q. 4	Solve Any Two of the following.				CO 3	12
A)	Describe advantages & disadvantages of conduit wiring?					6
B)	Explain nahani trap & gully with sketches?					6
C)	Explain Lumen method of lighting design?					6
Q.5	Solve Any Two of the following.				CO 4	12
A)	Describe working of air conditioning system?					6
B)	Explain natural ventilation with sketch?					6
C)	Describe wind effect & stack effect in ventilation?					6
Q. 6	Solve Any Two of the following.					12
A)	Explain Sabine's formula for acoustics?					6
B)	Write a short note on Firefighting measures?					6
C)	Describe reverberation of sound in acoustics?					6
	*** End ***					